USING INSTRUCTION TUNING TO CLASSIFY INFORMATION AS PRIVILEGED

NLP FINAL PROJECT

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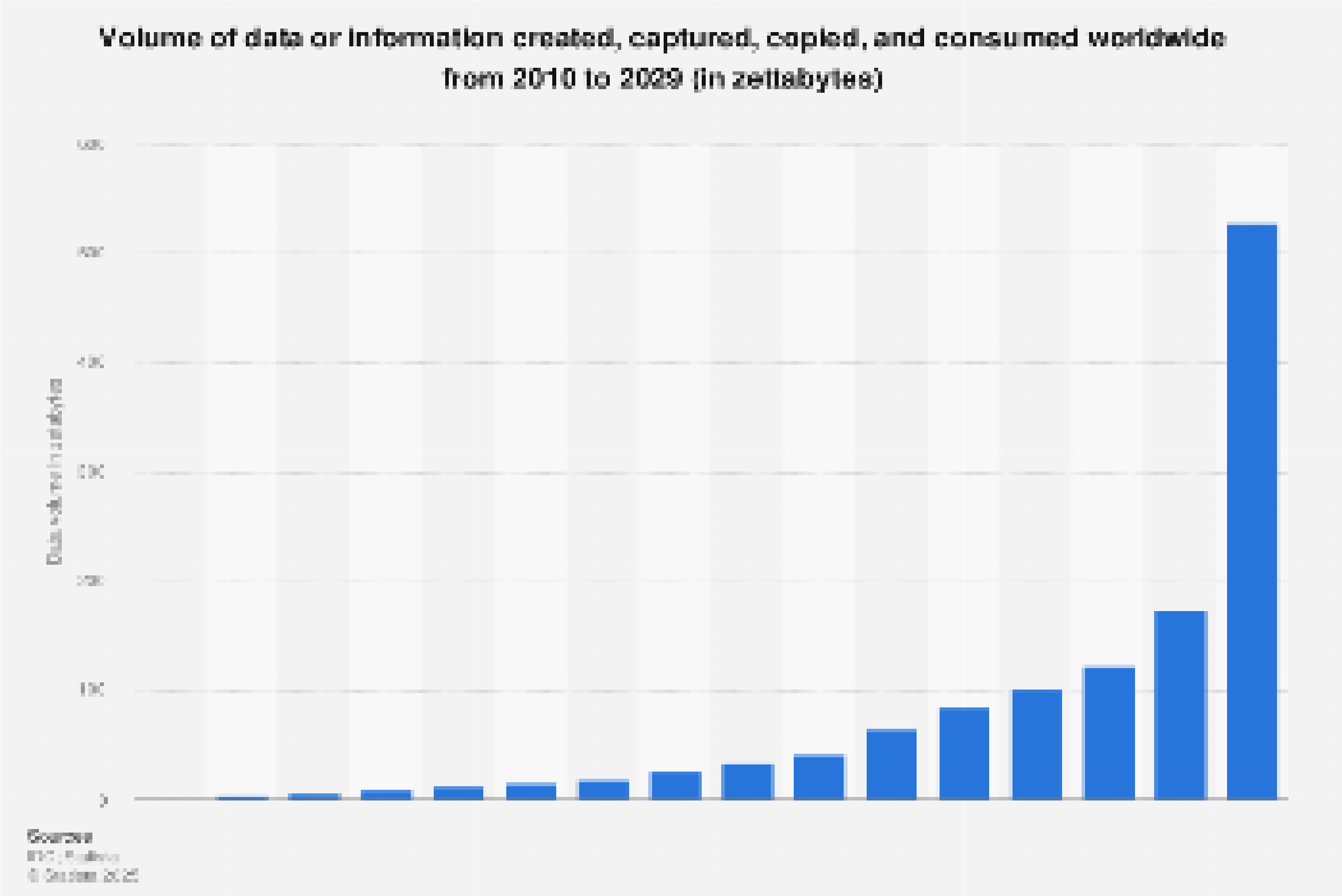
Electronically-Stored Information (ESI) is growing too rapidly to sustain. How can we replace a manual process with an automated/partially-automated (human-in-the-loop) process to handle this data?



Source: https://www.statista.com/statistics/871513/worldwide-data-created/

OVERVIEW

- What is privilege and how can we automate classification
- Building the Dataset
- Model Selection
- Results

PROBLEM

How can we identify privileged information?

What is privilege?

Confidential information/communication that is protected by law (i.e. attorney-client privilege)

Current approach for identifying privileged information requires manually reading/classifying each resource

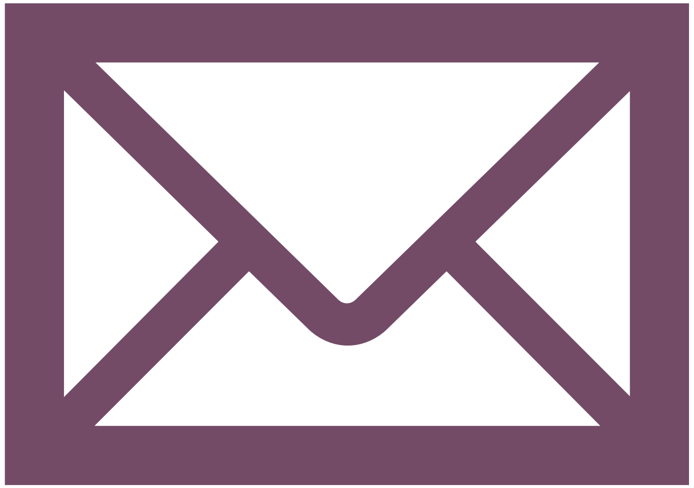
Not time efficient

Different people may classify the same document differently (subjective)

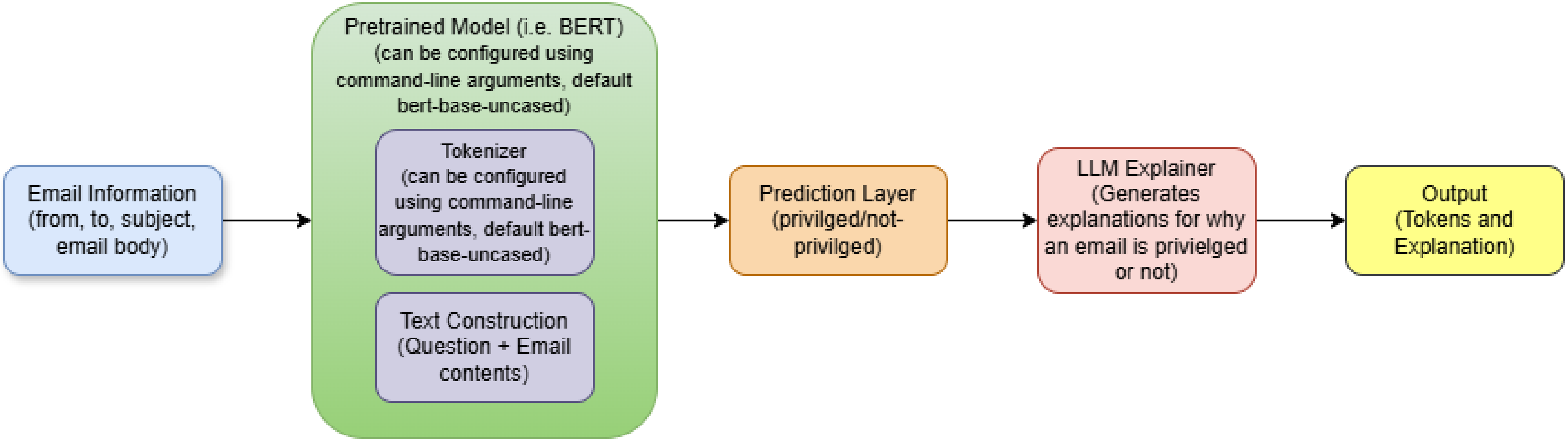
The amount of data that needs classified grows too quickly to sustain manual classification

DATASET

- Enron Dataset
  - Large, public corpus of approximately 500,000 real-world internal emails from ~150 senior Enron employees, released by the Federal Energy Regulatory Commission (FERC) after the company’s 2001 collapse. It is widely used in NLP, social network analysis, and AI training for machine learning.
- Weak Labeling
  - Used to label the original dataset as Privileged/Not Privileged

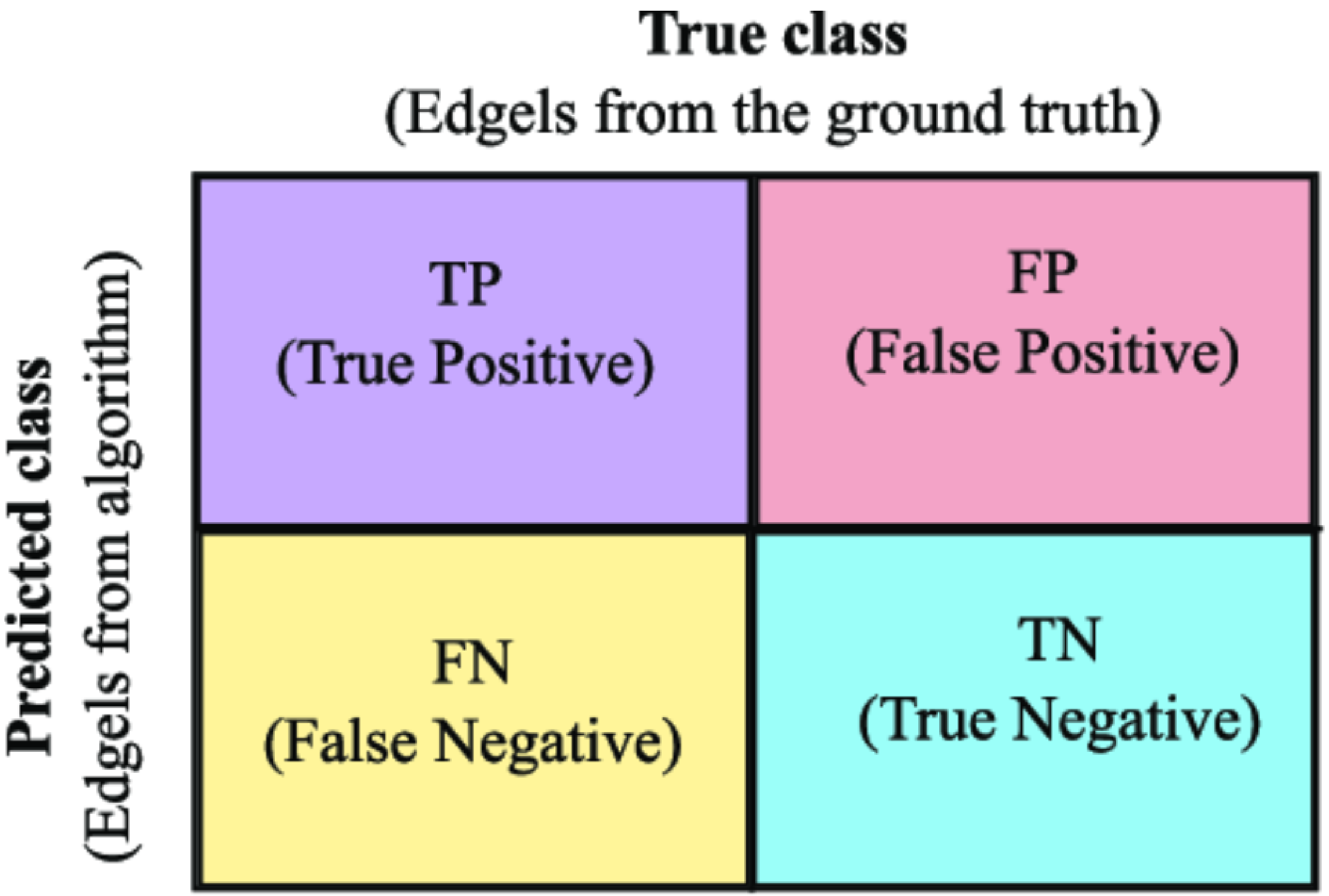


MODEL DIAGRAM



PREDICTION

- What does the prediction mean?
  - False Positive : Not privileged information classified as 'Privilege'
  - False Negative : Privileged information classified as 'Not Priviege' (legal risk)



Source: https://www.researchgate.net/figure/A-confusion-matrix-to-define-TP-TN-FP-and-FN_fig3_352393233

EXAMPLE

Input: From: john.doe@company.com, To: jane.smith@company.com, Subject: Q3 Budget Review, Body: Hi Jane, please review the attached budget for Q3. Let me know if you have any concerns before Friday.

Prediction: Not Privileged because The budget is for the following year.

Input: From: legal@company.com, To: ceo@company.com, Subject: Attorney-Client Privileged: Contract Review, Body: This communication is confidential and intended for legal advice. Please review the contract terms regarding liability and indemnification.

Prediction: Privileged because The email is confidential and intended for legal advice.

RESULTS

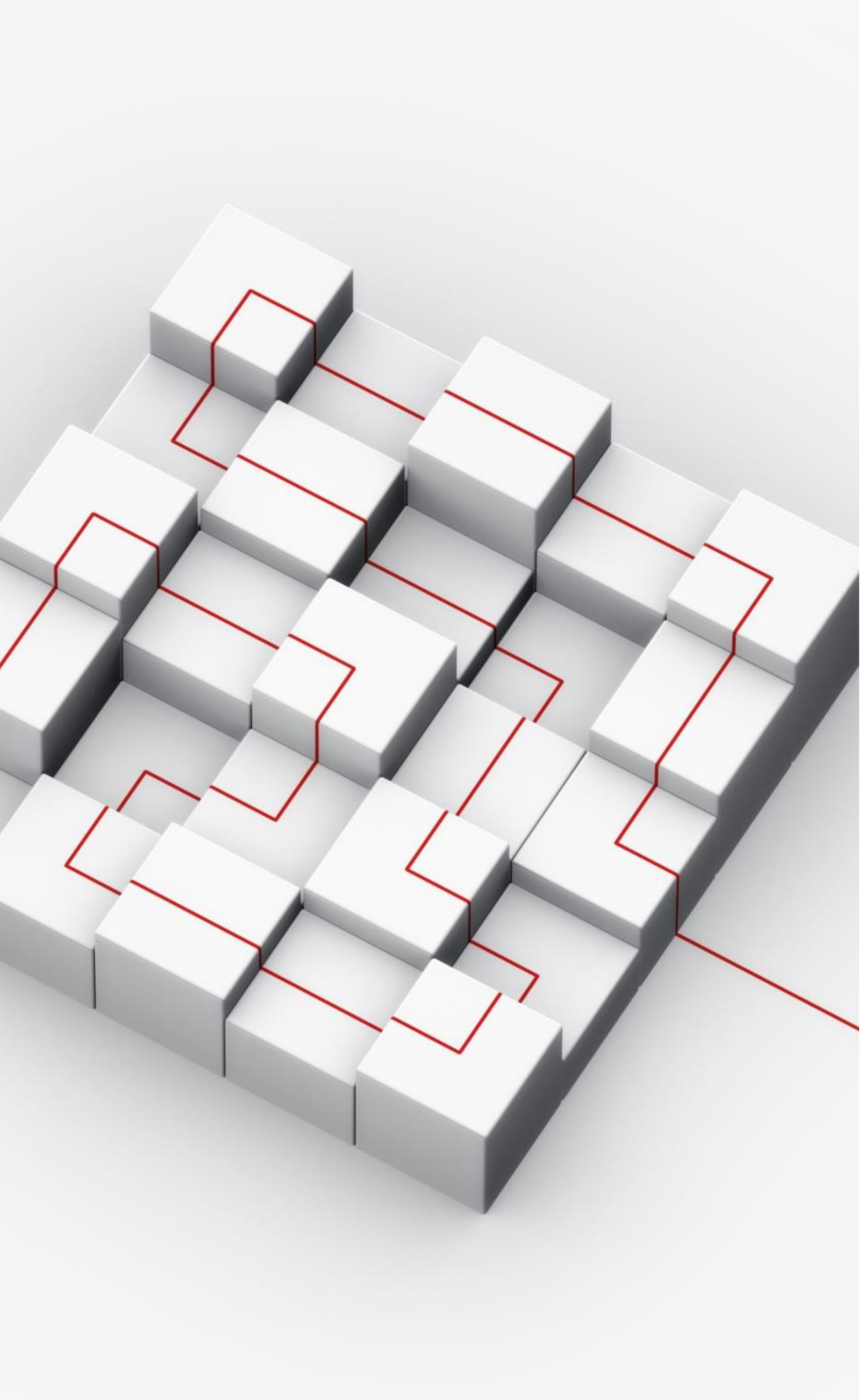
Experiment #	Hyperparameters Tuned	Accuracy	Precision	Recall	F1 Score
1	None (BERT base)	0.9620	0.8977	0.6311	0.7411
2	Max length 128	0.9470	0.7660	0.5545	0.6433
3	**Max length 512**	**0.9798**	**0.9558**	**0.8028**	**0.8726**
4	Train on 15k observations	0.9488	0.8302	0.5104	0.6322
5	Train on 45k observations	0.9714	0.9286	0.7239	0.8136
6	Learning Rate 1e-5	0.9576	0.8897	0.5800	0.7022
7	Learning Rate 3e-5	0.9570	0.8121	0.6520	0.7233
8	Batch Size 16	0.9600	0.8199	0.6868	0.7475
9	Epochs 2	0.9606	0.8634	0.6450	0.7384
10	RoBERTa model	0.9570	0.8803	0.5800	0.6993

WHAT NEXT?

Use a private dataset that includes actual privileged information within a specific field to train

Support a third classification that is "uncertain" that individuals would manually classify

Use RAG for explanation generation with a temperature that discourages creativity (low temperature -> fewer hallucinations)



SUMMARY

- Manually classifying Privileged information is unsustainable
- Privileged data is not publicly available
- Encoder-only model architecture for classification (i.e. BERT)
- Classified emails with an explanation for the prediction

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